

2025 December 2

FOOD INSECURITY & HEALTH OUTCOMES

Data Science I – Experience, Final Project

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Agenda

1 Introduction

2 Data and Cleaning

3 Analyses and Data Visualizations

Introduction

- Food insecurity impacts over 42 million Americans each year and is a risk factor for chronic disease and various mental health and well-being challenges
- SNAP benefits are standardized; meal costs are not

Main question: How do food insecurity rates and average annual shortfall per capita of SNAP benefits influence 2022 obesity and diagnosed diabetes rates in U.S. counties?

Data collection

All county-level data

Feeding America Map the Meal Gap, 2010- 2022

- Weighted annual food shortfall
- Food insecurity rates
- Cost per meal

Centers for Disease Control PLACES, 2022

- Diagnosed diabetes among adults rates
- Obesity among adults rates

American Community Survey, 2022

- Median household income
- Percent one-race, white only
- Percent high school and college education attainment
- Percent over 65 years of age
- Percent household without employment

Cleaning and processing

Steps:

- Merging the datasets using FIPS code.
- Assessment of data quality:
 - + Check duplicates, handle NaN values.
 - + Ensure variables from Map the meal gap, Health prevalence, and Census datasets across time and county.
- Create new indicators:
 - + Short-term, medium-term and long-term exposures indicator.
 - + Shortfall per capita.
- Standardize variables for consistency and meaningful interpretation.

year	number_of_counties
2010	3143
2011	3143
2012	3143
2013	3143
2014	3142
2015	3142
2016	3142
2017	3142
2018	3141
2019	3142
2020	3143
2021	3143
2022	3144

Variable	Missing_count	Missing_per
high_threshold_in_state	0	0.00
per_fi_less_or_equal_to_low_threshold	0	0.00
county_name	0	0.00
food_insecurity_rate	0	0.00
per_fi_children_hh_income_above_185_fpl	10	0.02
fips	0	0.00
low_threshold_type	0	0.00
num_of_food_insecure_persons	0	0.00
cost_per_meal	0	0.00
child_food_insecurity_rate	7	0.02
num_of_food_insecure_children	5	0.01
high_threshold_type	0	0.00
per_fi_children_hh_income_below_185_fpl	10	0.02
state_name_raw	0	0.00
low_threshold_in_state	0	0.00
state_name	0	0.00
weighted_annual_shortfall	0	0.00
per_fi_btwn_thresholds	17363	42.50
per_fi_greater_high_threshold	0	0.00
year	0	0.00

Analyses

Basic analysis

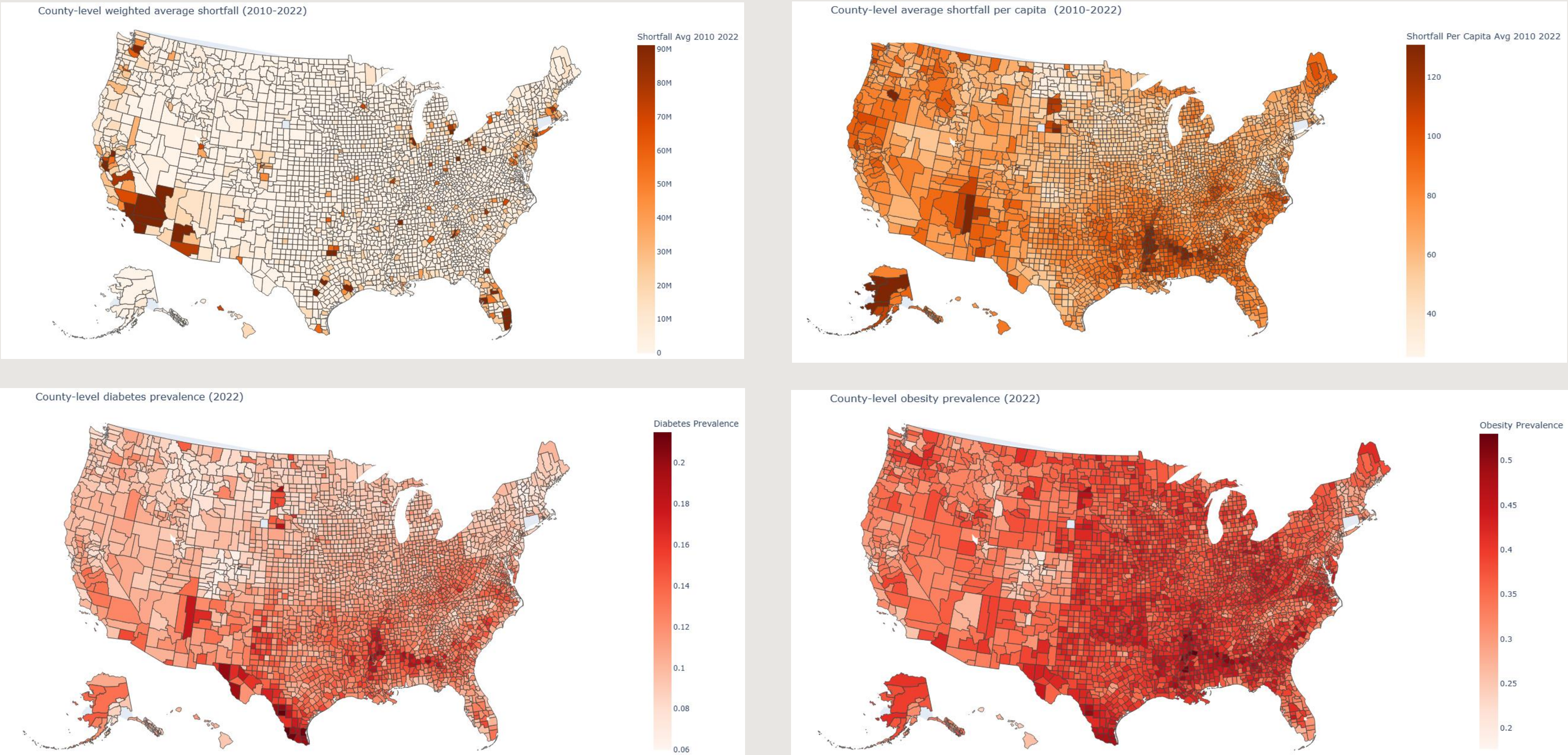


Figure 1: (a) County-level Average shortfall (2010-2022), (b) County-level Average shortfall per capita (2010-2022) (c.) County-level diabetes prevalence in 2022, (d) County-level obesity prevalence in 2022. Shortfall per capita seems to be a more informative indicator.

Basic analysis (cont'd)

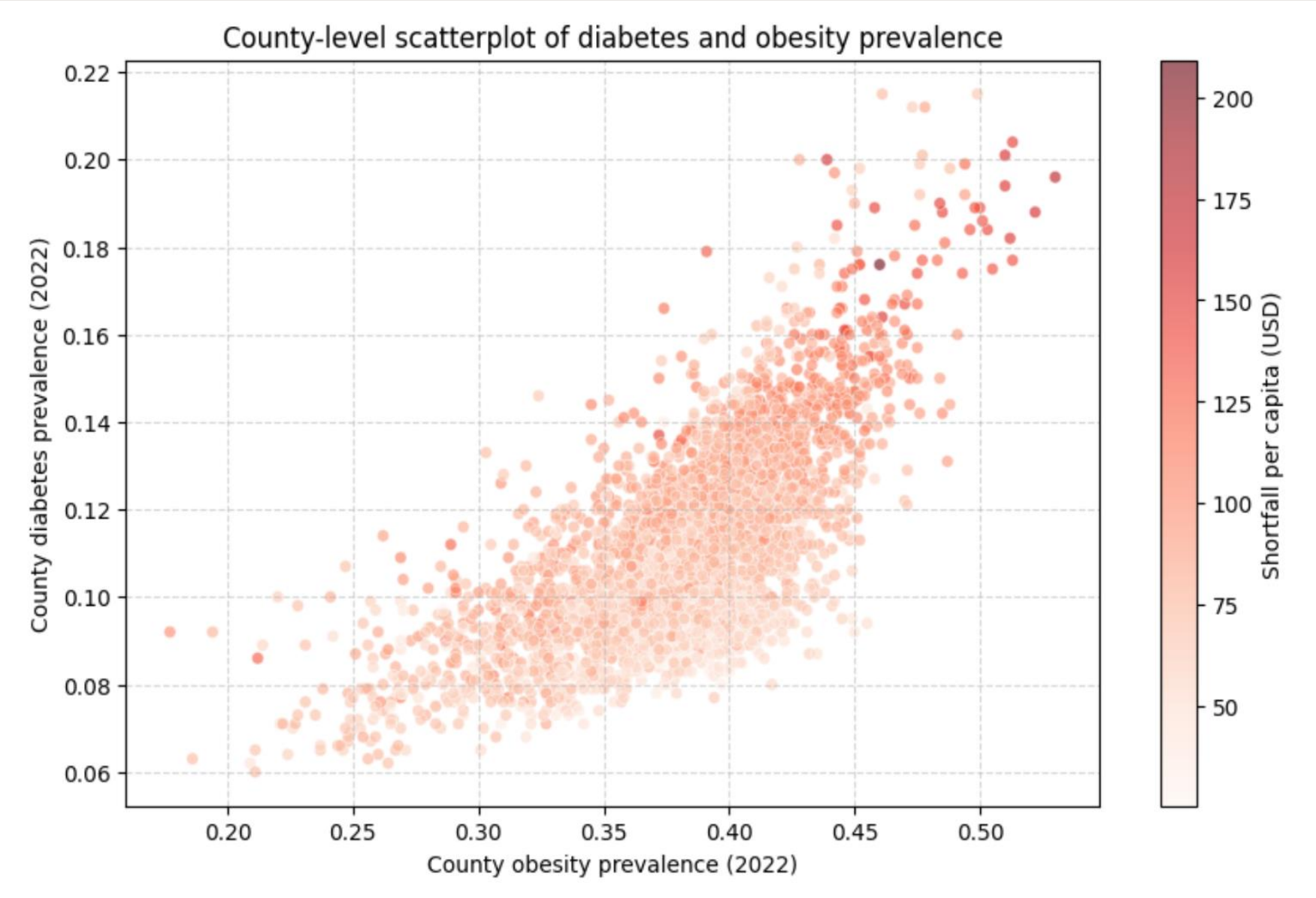


Figure 2: County-level scatterplot of diabetes and obesity prevalence. Darker points (higher FI shortfall per capita) tend to be on the higher end of diabetes and obesity prevalence percentages.

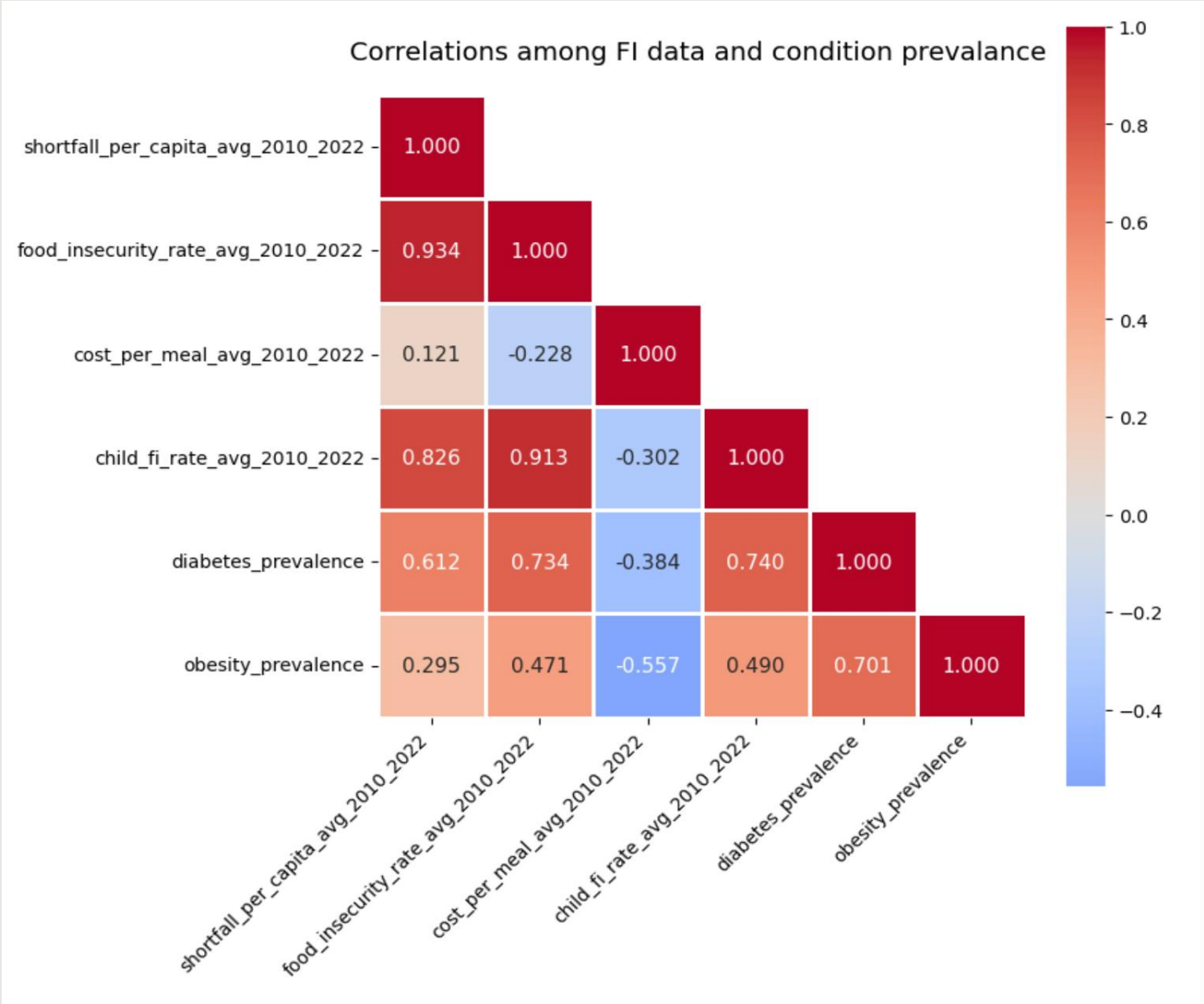


Figure 3: Heatmap of correlation matrix among FI indicators. Counties with higher food insecurity also have higher food budget shortfalls. FI rate (or Shortfall per capita) correlated with diabetes prevalence more than obesity prevalence. Cost per meal have negative correlation both with FI indicators and condition outcomes.

Descriptive analysis – *What is the relationship between food insecurity rates over time and diagnosed diabetes and obesity rates?*



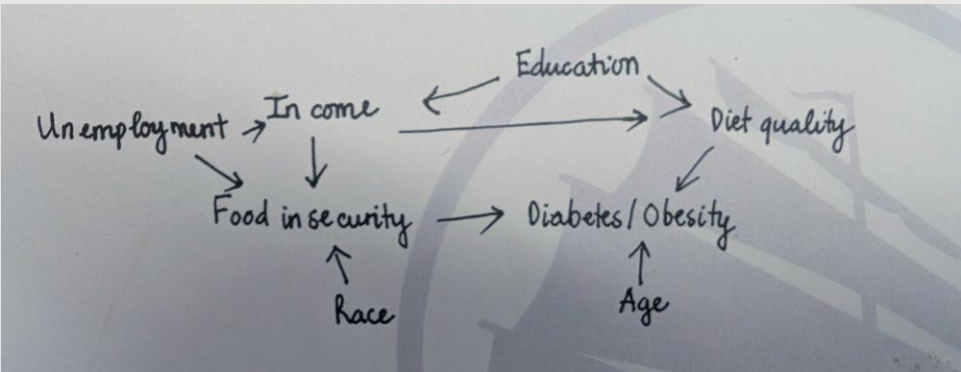
Explanatory analysis

Steps:

- Draw the casual diagram.
- Fit OLS models to quantify relationships between FI indicators and condition prevalence.
- Adjust for the socioeconomic factors: income, race, age, education, and unemployment.

Results:

- Food insecurity indicators show a positive and statistically significant association with diabetes and obesity rates. Most socioeconomic factors behave accordingly to our expectations (income, race, education).
- Good R^2 values indicate that the model we build have some explanatory power.



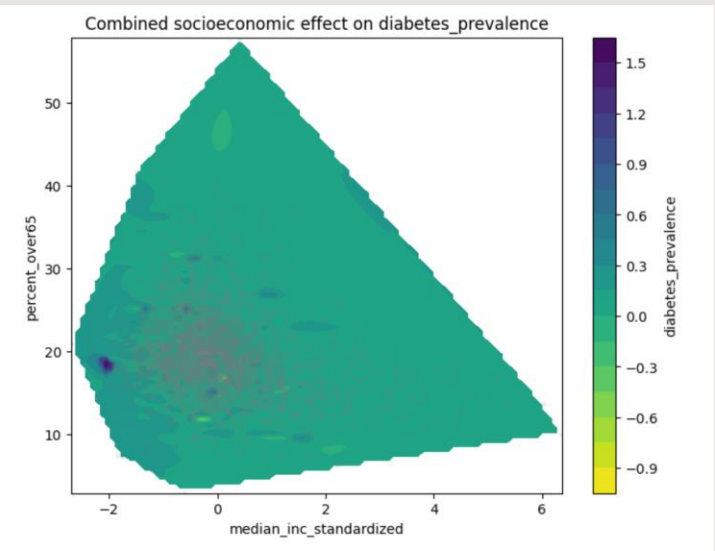
$$\text{Diabetes Prevalence} = \beta_0 + \beta_1 \cdot \text{Avg Shortfall Per Capita} + \beta_2 \cdot \text{Median Income Statardized} + \beta_3 \cdot \text{Per Col Grad} + \beta_4 \cdot \text{Per Families w/o Employment} + \beta_5 \cdot \text{Per Over 65} + \beta_6 \cdot \text{Per White Only} + \epsilon$$

OLS Regression Results						
=====						
Dep. Variable:	diabetes_prevalence	R-squared:	0.725			
Model:	OLS	Adj. R-squared:	0.724			
Method:	Least Squares	F-statistic:	1376.			
Date:	Sat, 29 Nov 2025	Prob (F-statistic):	0.00			
Time:	19:45:57	Log-Likelihood:	9412.8			
No. Observations:	3143	AIC:	-1.881e+04			
Df Residuals:	3136	BIC:	-1.877e+04			
Df Model:	6					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	0.1846	0.002	78.520	0.000	0.180	0.189
shortfall_per_capita_avg_2010_2022	4.382e-05	1.92e-05	2.283	0.022	6.19e-06	8.14e-05
median_inc_standardized	-0.0076	0.000	-18.320	0.000	-0.008	-0.007
percent_families_wo_employment	0.0006	5.46e-05	11.259	0.000	0.001	0.001
per_col_grad	-0.0006	3.19e-05	-20.115	0.000	-0.001	-0.001
percent_white_only	-0.0006	1.69e-05	-35.227	0.000	-0.001	-0.001
percent_over65	-0.0013	6.25e-05	-20.521	0.000	-0.001	-0.001
=====						

OLS Regression Results						
=====						
Dep. Variable:	diabetes_prevalence	R-squared:	0.714			
Model:	OLS	Adj. R-squared:	0.713			
Method:	Least Squares	F-statistic:	1563.			
Date:	Mon, 01 Dec 2025	Prob (F-statistic):	0.00			
Time:	15:39:24	Log-Likelihood:	9350.6			
No. Observations:	3143	AIC:	-1.869e+04			
Df Residuals:	3137	BIC:	-1.865e+04			
Df Model:	5					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	0.1811	0.002	76.204	0.000	0.176	0.186
shortfall_per_capita_avg_2010_2022	0.0001	1.8e-05	7.124	0.000	9.3e-05	0.000
median_inc_standardized	-0.0080	0.000	-19.024	0.000	-0.009	-0.007
per_col_grad	-0.0007	3.23e-05	-21.435	0.000	-0.001	-0.001
percent_over65	-0.0009	5.19e-05	-16.839	0.000	-0.001	-0.001
percent_white_only	-0.0006	1.71e-05	-33.422	0.000	-0.001	-0.001
=====						



Explanatory analysis

1 Obesity rate ~ Food Insecurity Rate (FIR)

OLS Regression Results						
=====						
Dep. Variable:	obesity_rate		R-squared:	0.222		
Model:	OLS		Adj. R-squared:	0.221		
Method:	Least Squares		F-statistic:	891.2		
Date:	Sun, 30 Nov 2025		Prob (F-statistic):	1.58e-172		
Time:	19:11:07		Log-Likelihood:	5566.0		
No. Observations:	3130		AIC:	-1.113e+04		
Df Residuals:	3128		BIC:	-1.112e+04		
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	0.2957	0.003	102.203	0.000	0.290	0.301
fi_mean_10_22	0.6022	0.020	29.853	0.000	0.563	0.642

2 Obesity rate ~ Food Insecurity Rate (FIR) + SES Factors

OLS Regression Results					
=====					
Dep. Variable:	obesity_rate	R-squared:	0.434		
Model:	OLS	Adj. R-squared:	0.433		
Method:	Least Squares	F-statistic:	479.3		
Date:	Sun, 30 Nov 2025	Prob (F-statistic):	0.00		
Time:	19:11:07	Log-Likelihood:	6064.7		
No. Observations:	3130	AIC:	-1.212e+04		
Df Residuals:	3124	BIC:	-1.208e+04		
Df Model:	5				
Covariance Type:	nonrobust				
=====					
	coef	std err	t	P> t	[0.025

const	0.6474	0.014	46.243	0.000	0.620
fi_mean_10_22	-0.0696	0.030	-2.334	0.020	-0.128
median_household_income	-1.684e-06	6.23e-08	-27.033	0.000	-1.81e-06
per_hs_grad	-0.1235	0.014	-8.790	0.000	-0.151
per_white_only	-0.0096	0.005	-1.983	0.047	-0.019
per_over65	-0.1788	0.015	-11.909	0.000	-0.208

3 Obesity rate ~ Food Insecurity Rate (FIR) + SES Factors + State Fixed Effects

OLS Regression Results			
=====			
Dep. Variable:	obesity_rate	R-squared:	0.750
Model:	OLS	Adj. R-squared:	0.746
Method:	Least Squares	F-statistic:	170.8
Date:	Sun, 30 Nov 2025	Prob (F-statistic):	0.00
Time:	19:11:07	Log-Likelihood:	7342.7
No. Observations:	3130	AIC:	-1.458e+04
Df Residuals:	3075	BIC:	-1.424e+04
Df Model:	54		
Covariance Type:	nonrobust		

Predictive analysis - *Can we predict county-level diabetes (or obesity) prevalence using shortfall per capita and other socioeconomic factors?*

Diabetes rate ~ Shortfall per capita + Median Inc + Race + Age + Education

Steps:

- Predictors: shortfall per capita + socioeconomic factors
- Perform CV to find the best hyper parameter alpha. Fit Ridge and LASSO models.
- Evaluate which predictors matter the most.

	Model	Best alpha	Test R^2 Score	Test RMSE
:---	:-----	:-----	:-----	:-----
0	Ridge	10	64.79%	1.33%
1	LASSO	0.0001	64.81%	1.33%

```
Optimized Lasso coefficients for diabetes prevalence
shortfall_per_capita_avg_2010_2022    0.002799
median_inc_standardized               -0.007293
percent_white_only                    -0.009935
percent_over65                       -0.004079
per_col_grad                         -0.006937
dtype: float64
```

```
Optimized Lasso coefficients for obesity prevalence
shortfall_per_capita_avg_2010_2022   -0.006322
median_inc_standardized              -0.018381
percent_white_only                   -0.008087
percent_over65                      -0.007837
per_col_grad                        -0.020650
dtype: float64
```

Results :

- Both model relatively explain partially amount of the variance in diabetes prevalence. Both have low RMSE.
- Shortfall per capita – weak predictor yet still provide predictive information that either Race demography or Education alone can not capture.

Predictive analysis – *Can we predict whether a county will have a diabetes rate higher than the U.S. average based on its food insecurity rate and socioeconomic factors?*

Steps:

- Select predictors: mean food insecurity rate 2010-2022, median HH income, age, race, HS education attainment
- Perform **Random Forest Classification**
- Evaluate confusion matrix metrics

Results: Diabetes Rates

TN: 305	FP: 42
FN: 45	TP: 234

Accuracy: 0.86

Precision: 0.85

Recall/Specificity: 0.84

Importance of false
positives and false
negatives

Thanh's FI interested variable: Shortfall per capita | Julia's FI interested variable: FI rate

Reflections

Reflections

- Model for Regression problem assume linearity. We could try Random Forest or XGBoost to capture nonlinearity. Similarly, for the Classification, we started with Random Forest because of interpretability but should do model comparison with XGBoost .
- Explore in more detail the effect of time-lag between food insecurity exposure and prevalence of health outcomes.
- Add more relevant predictors in our model (urban/rural indicator, % uninsured, access to hospitals/clinics, etc.)
- Counties are not independent. We could try spatial regression models to adjust for regional clustering.

Questions?

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THANK YOU!

Thanh Tran Julia Esposito